

All-Optical Radiation Reaction at 10^{21} W/cm^2

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Notebook: Óscar Amaro, November 2022 @ GoLP-EPP

Introduction

In this notebook we reproduce figure 3.

Figure 3: Electron beam energy loss

(In the units chosen in the paper, the classical electron radius, present in the Thompson cross section, is $r_e = \frac{e^2}{m c^2}$ while in SI $r_e = \frac{1}{4 \pi \epsilon_0} \frac{e^2}{m c^2}$)

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In[ ]:= (* LP PW laser colliding against a relativistic electron *)
Clear[γ, γf, γ0, k, k3, η, EGeV, τ0, τ0fs, e, m, c, λ, a0, ω0, ΔEE, χ, ΔEEχ, ε0]

(* equation 3 *)
γf = γ0 / (1 + k γ0); (*[] final electron energy *)
k3 =  $\frac{1}{4 \pi \epsilon_0} (1 - \cos[\pi])^2 \frac{\eta}{3} \frac{e^2 \omega^2}{m c^3} a_0^2 \tau_0$ ; (*[] CRR k factor from equation 3 *)
k = (1 - Cos[π])2 3.2 × 10-5 (1 10-22) τ0fs;
(*[] CRR k factor Engineering formula *)
ΔEE = (γ0 - γf) / γ0; (*[] 0≤energy loss≤1 *)

(* χ=1: χ~2 γ0 a0/aS *)
(*Solve[γf==γ0/(1+k γ0), γ0][[1,1,2]]*)
γ = Solve[1 == 2 γ a0 / (411823), γ][[1, 1, 2]];
ΔEEχ = ΔEE /. {γ0 → γ};

(* physical constants *)
e = 1.6 × 10-19; (*[C]*)
m = 9.1 × 10-31; (*[Kg]*)
c = 3 × 108; (*[m/s]*)
ε0 = 8.854 × 10-12; (*[F/m] vacuum permittivity*)

(* parameters *)
η = 0.4; (*[] temporal profiles *)
τ0 = 26.5 × 10-15; (*[s]*)
τ0fs = τ0 1015; (*[fs]*)
λ = 1; (*[μm] laser central wavelength *)
ω0 = 2 π c / (λ 10-6) // N; (*[1/s] laser frequency *)
a0 = 0.855 Sqrt[1 10-18] λ; (*[] laser a0 *)
γ0 = EGeV / (0.511 × 10-3); (*[] electron γ from GeV to boost *)

(* compare the CRR k factors of equation 3 and the "engineering formula" *)
k3
k
(*k ~ 4.64 10-7 a02 for these parameters *)

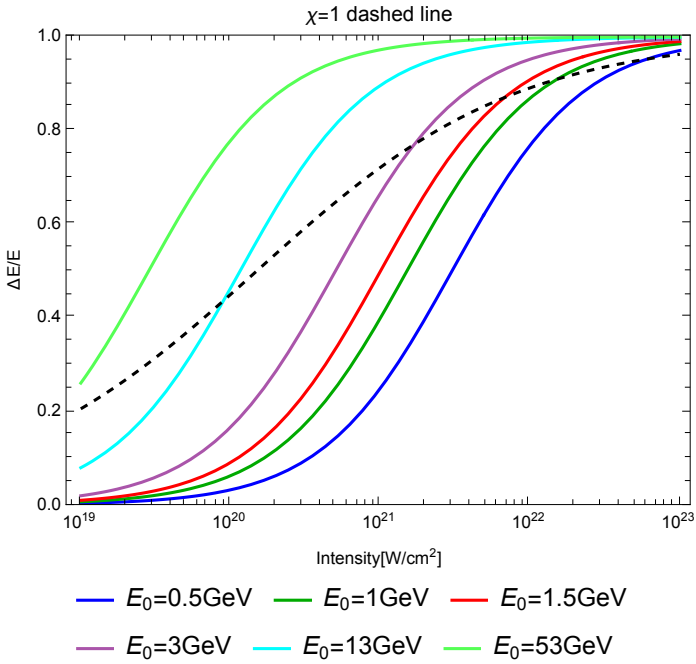
(* plot *)
LogLinearPlot[{ΔEE /. {EGeV → 0.5}, ΔEE /. {EGeV → 1}, ΔEE /. {EGeV → 1.5},
  ΔEE /. {EGeV → 3}, ΔEE /. {EGeV → 13}, ΔEE /. {EGeV → 53}, ΔEEχ},
  {1, 1019, 1023}, Frame → True, FrameLabel → {"Intensity[W/cm2]", "ΔE/E"},
  PlotRange → {0, 1}, PlotStyle → {Blue, Darker[Green], Red,
  Lighter[Purple], Cyan, Lighter[Green], {Dashed, Black}}, PlotLegends →
  {"E0=0.5GeV", "E0=1GeV", "E0=1.5GeV", "E0=3GeV", "E0=13GeV", "E0=53GeV"},
  AspectRatio → 3 / 4, PlotLabel → "χ=1 dashed line"]

```

Out[]=

3.43767 × 10⁻²⁵ 1

Out[*]=
 $3.392 \times 10^{-25} \text{ I}$
Out[*]=



η factor for the two envelopes (+Gaussian)

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Clear[t,  $\tau$ ,  $\tau_0$ , aPoly, aSin2, aGauss, intPoly, intSin2, intGauss]
 $\tau_0 = 1$ ; (*[] normalize all times to  $\tau_0$  *)
 $\tau = t / \tau_0$ ;

aPoly =  $10 \tau^3 - 15 \tau^4 + 6 \tau^5$ ; (* polynomial envelope. instead of using  $\tau =$ 
 $\sqrt{2} t / \tau_0$  as in the paper (which would give incorrect integrals) we use
    simply  $\tau = t / \tau_0$  to be able to compare between the two envelopes *)
aSin2 = Sin[ $\frac{\pi t}{2 \tau_0}$ ]2; (* sin2 envelope *)
aGauss = Exp[-4 Log[2] (1 -  $\frac{t}{\tau_0}$ )2]; (* sin2 envelope *)

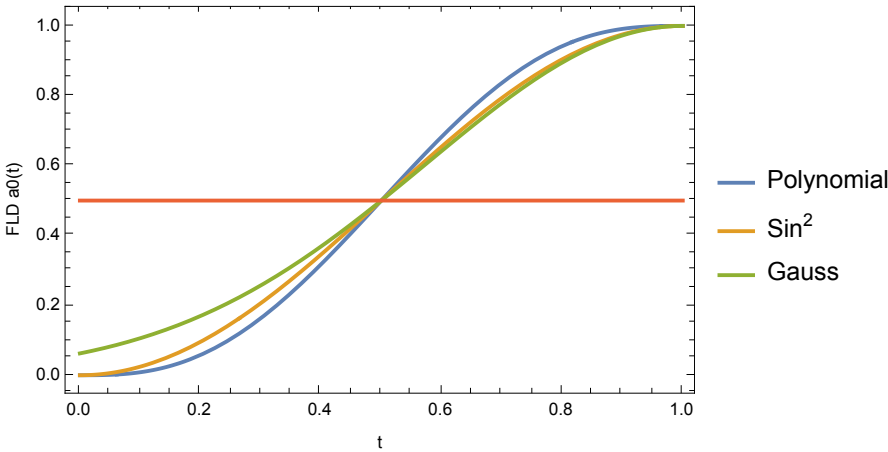
(* plot the three envelopes with same fwhm in field, not intensity *)
Plot[{aPoly, aSin2, aGauss, 0.5}, {t, 0,  $\tau_0$ }, Frame → True,
    FrameLabel → {"t", "FLD a0(t)"}, PlotLegends → {"Polynomial", "Sin2", "Gauss"}]
(* plot also intensity *)
Plot[{aPoly2, aSin22, aGauss2, 0.5}, {t, 0,  $\tau_0$ },
    Frame → True, FrameLabel → {"t", "Intensity a0(t)2"},
    PlotLegends → {"Polynomial", "Sin2", "Gauss"}]

(* the k factor will be proportional to a(t)2. since the
    envelopes are symmetric, we only integrate up to t= $\tau_0$  *)
intPoly = Integrate[aPoly2, {t, 0,  $\tau_0$ }]
Print["poly: ", intPoly, "=", intPoly // N]
intSin2 = Integrate[aSin22, {t, 0,  $\tau_0$ }]
Print["sin2: ", intSin2, "=", intSin2 // N]
intGauss = Integrate[aGauss2, {t, - $\infty$ ,  $\tau_0$ }]
Print["gauss: ", intGauss, "=", intGauss // N]

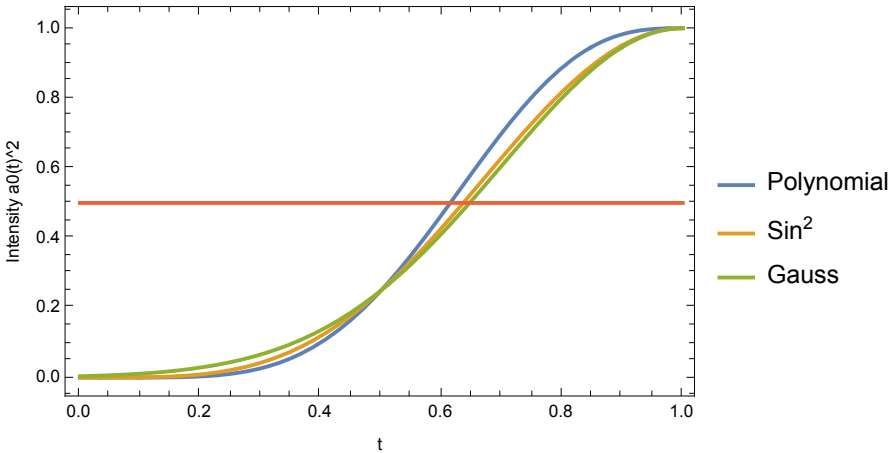
(* the ratio of the integrals from this notebook *)
intSin2 / intPoly // N
(* the ratio of the integrals from the paper *)
0.375 / 0.392

```

Out[]=



Out[]=



poly: $\frac{181}{462} = 0.391775$

sin2: $\frac{3}{8} = 0.375$

gauss: $\frac{1}{4} \sqrt{\frac{\pi}{\text{Log}[4]}} = 0.376346$

Out[]=

0.957182

Out[]=

0.956633